Gaussian fixed point

$t^* = 0$, $u^* = 0$

Write $t = t^* + \delta t$, $u = u^* + \delta u$ and expand (Linearize) in $\delta t, \delta u$:

$$\frac{d}{dl}(\delta t) = 2\,\delta t + \left(\frac{6 K_d \Lambda^{d-2}}{\rho}\right)\delta u$$

Since $t^* = 0$ in denominator

$$\frac{d}{dl}(\delta u) = (4-d)\,\delta u$$

or, $\frac{d}{dl}\vec{v}(l) = M\,\vec{v}(l)$

$$\vec{v}(l) = \begin{pmatrix}\delta t \\ \delta u\end{pmatrix} \quad , M = \begin{pmatrix} 2 & a \\ 0 & 4-d \end{pmatrix} \qquad a \equiv \frac{6 K_d \Lambda^{d-2}}{\rho}$$

Eigenvalues of M: $\begin{vmatrix} 2-\lambda & a \\ 0 & 4-d-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 = 2,\ \lambda_2 = 4-d$

(do not depend on a)

Eigenvectors: $\begin{pmatrix} 2 & a \\ 0 & 4-d \end{pmatrix}\vec{e}_1 = \lambda_1\,\vec{e}_1 = 2\vec{e}_1$

$$\vec{e}_1: \quad \begin{cases} 2\,e_{1,1} + a\,e_{1,2} = 2\,e_{1,1} \Rightarrow e_{1,1} = 1 \\ (4-d)\,e_{1,2} = 2\,e_{1,2} \Rightarrow e_{1,2} = 0 \end{cases}$$

$$\vec{e}_1 = \begin{pmatrix}1\\0\end{pmatrix}$$

$$\begin{pmatrix} 2 & a \\ 0 & 4-d \end{pmatrix}\vec{e}_2 = (4-d)\,\vec{e}_2$$

$$\vec{e}_2: \quad \begin{cases} 2\,e_{2,1} + a\,e_{2,2} = (4-d)\,e_{2,1} \\ (4-d)\,e_{2,2} = (4-d)\,e_{2,2} \end{cases}$$

$$(2-d)\,e_{2,1} = +a\,e_{2,2}$$

$$\Rightarrow \quad e_{2,1} = -a, \quad e_{2,2} = d-2$$

$$\vec{e}_2 = \begin{pmatrix} -a \\ d-2 \end{pmatrix}$$

Then $\vec{v}(\ell) = v_1(\ell)\,\vec{e}_1 + v_2(\ell)\,\vec{e}_2$

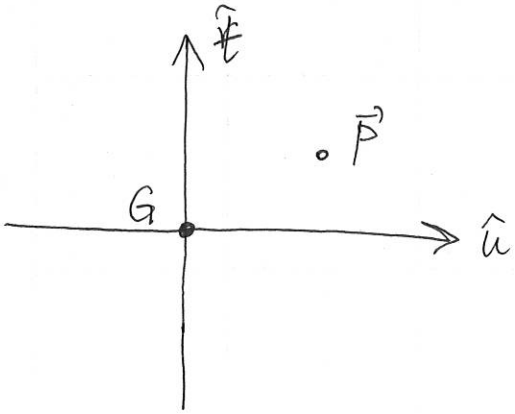
$$\frac{d}{d\ell}\vec{v}(\ell) = M\,\vec{v}(\ell) = M\left(v_1(\ell)\,\vec{e}_1 + v_2(\ell)\,\vec{e}_2\right) =$$

$$= v_1(\ell)\,\underbrace{M\vec{e}_1}_{\lambda_1\vec{e}_1} + v_2(\ell)\,\underbrace{M\vec{e}_2}_{\lambda_2\vec{e}_2}$$

$$\Rightarrow \frac{dv_1}{d\ell}\,\vec{e}_1 + \frac{dv_2}{d\ell}\,\vec{e}_2 = v_1\lambda_1\,\vec{e}_1 + v_2\lambda_2\,\vec{e}_2$$

$$\Rightarrow \frac{dv_1}{d\ell} = \lambda_1 v_1 \Rightarrow v_1(\ell) = v_1(0)\,e^{\lambda_1\ell} = v_1(0)\,e^{2\ell}$$

$$\frac{dv_2}{d\ell} = \lambda_2 v_2 \Rightarrow v_2(\ell) = v_2(0)\,e^{\lambda_2\ell} = v_2(0)\,e^{(4-d)\ell}$$



$$P: \; t\,\hat{t} + u\,\hat{u} = \vec{P}$$

$$\hat{e}_1 = \hat{t}$$

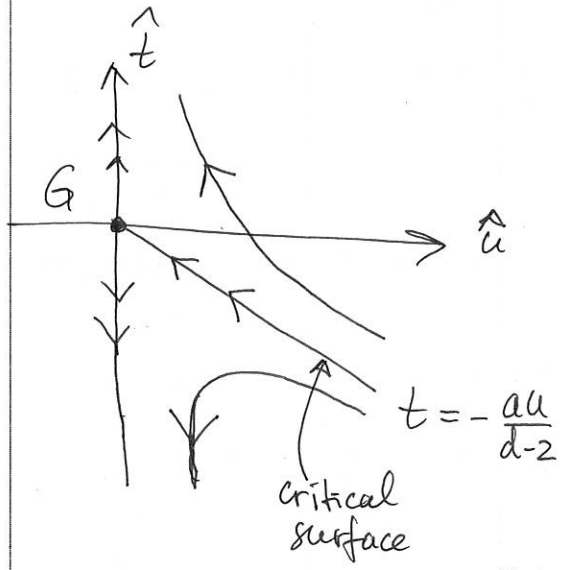
$$\hat{e}_2 = -a\hat{t} + (d-2)\hat{u}$$

$$\hat{u} = \frac{\hat{e}_2 + a\hat{e}_1}{d-2}$$

$$\Rightarrow \vec{P} = t\,\hat{e}_1 + \frac{u}{d-2}\left(\hat{e}_2 + a\hat{e}_1\right)$$

$$= \left(t + \frac{ua}{d-2}\right)\hat{e}_1 + \frac{u}{d-2}\,\hat{e}_2$$

$\boxed{D>4}$ (unstable) relevant direction $\hat{e}_1$
(stable) irrelevant direction $\hat{e}_2$ = critical surface



Fixed point at $t + \frac{ua}{d-2} = 0$

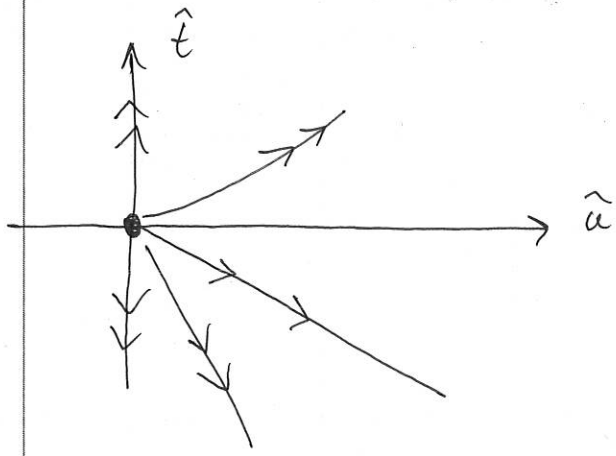
$$t = -\frac{a}{d-2} u$$

$\Rightarrow$ critical temperature is reduced by interaction u.

RG flow:

$$\vec{p}(\ell) = \left(t_0 + \frac{u_0 a}{d-2}\right) e^{2\ell} \hat{e}_1 + \frac{u_0}{d-2} e^{(4-d)\ell} \hat{e}_2 \xrightarrow[\ell\to\infty]{} \left(t_0 + \frac{u_0 a}{d-2}\right) e^{2\ell} \hat{e}_1$$

$\boxed{D<4}$ $\lambda_2 = 4-d > 0 \Rightarrow \hat{e}_2$ becomes unstable direction too!
(becomes relevant too) $\Rightarrow$ no phase transition



$$\vec{p}(\ell) \to \left(t_0 + \frac{u_0 a}{d-2}\right) e^{2\ell} \hat{e}_1 + \frac{u_0}{d-2} e^{(4-d)\ell} \hat{e}_2$$

No critical surface!
Run-away flow in all directions.

Kardar, ch. 5 RG up to u^2 order

$$\begin{cases} \dfrac{dt}{dl} = 2t + \dfrac{4u(n+2)K_d\Lambda^d}{t+\rho\Lambda^2} \\[2ex] \dfrac{du}{dl} = (4-d)u - \dfrac{4(n+8)K_d\Lambda^d}{(t+\rho\Lambda^2)^2}u^2 \end{cases}$$

(equ. 5.54)
n = number of components in vector field $\vec{\phi}$.

Fixed point $\dfrac{dt^*}{dl} = \dfrac{du^*}{dl} = 0$ (1) Gaussian, $u^*=0$, $t^*=0$ (Same as before)

(2) Wilson-Fisher (Heisenberg) "Critical exponents in 3.99 dimensions"

Set $4-d = \varepsilon \ll 1$ and expand to first order in ε:

$$u^* = \frac{\varepsilon(t^*+\rho\Lambda^2)^2}{4(n+8)K_d\Lambda^d} = \frac{\varepsilon\,\rho^2\Lambda^4}{4(n+8)K_d\Lambda^d} = \frac{\rho^2}{4(n+8)K_d}\varepsilon$$

set $d=4$ here because the result is already $O(\varepsilon)$.

$$t^* = -\frac{2u^*(n+2)K_d\Lambda^d}{t^*+\rho\Lambda^2} = -\frac{2(n+2)K_d\Lambda^d}{}\left(\frac{\rho^2\varepsilon}{4(n+8)K_d}\right)\cdot$$

$$\cdot\frac{1}{t^*+\rho\Lambda^2} = -\frac{(n+2)\rho\Lambda^2}{2(n+8)}\varepsilon$$

have to set $t^*=0$ here, already have ε in the numerator!

Now linearize : $t = t^* + \delta t$, $u = u^* + \delta u$

$$\Rightarrow \quad \frac{d}{dl}(\delta t) = 2(t^* + \delta t) + \frac{4(u^* + \delta u)(n+2) K_d \Lambda^d}{t^* + \rho\Lambda^2 + \delta t} =$$

$$= 2t^* + 2\delta t + \frac{4u^*(n+2) K_d \Lambda^d}{(t^* + \rho\Lambda^2)} \cdot \left(1 - \frac{\delta t}{t^* + \rho\Lambda^2}\right) +$$

$$+ \frac{4(n+2) K_d \Lambda^d}{t^* + \rho\Lambda^2} \delta u =$$

$$= \left(2t^* + \frac{4u^*(n+2) K_d \Lambda^d}{t^* + \rho\Lambda^2}\right) + \left(2 - \frac{4u^*(n+2) K_d \Lambda^d}{(t^* + \rho\Lambda^2)^2}\right)\delta t$$

$$\boxed{\frac{d}{dl}(t^*) = 0} \qquad + \frac{4(n+2) K_d \Lambda^d}{\rho\Lambda^2\left[1 - \frac{(n+2)\varepsilon}{2(n+8)}\right]} \delta u$$

$$\frac{d}{dl}(\delta t) = \left\{2 - 4(n+2) K_d \Lambda^d \left[\frac{\rho^2 \varepsilon}{4(n+8) K_d}\right] \cdot \frac{1}{\rho^2 \Lambda^4}\right\} \delta t + (\ldots)\delta u$$

$$= \left\{2 - \frac{(n+2)}{n+8}\right\} \delta t + a\,\delta u$$

$$\frac{d}{dl}(\delta u) = \varepsilon(u^* + \delta u) - \frac{4(n+8)K_d \Lambda^d}{(t^* + \rho\Lambda^2 + \delta t)^2} \underbrace{(u^* + \delta u)^2}_{\substack{(u^*)^2 + 2u^*\delta u \\ \Downarrow \\ O(\varepsilon^2)}}$$

$$= \varepsilon u^* - \frac{4(n+8)K_d \Lambda^d}{\rho^2 \Lambda^4} \frac{2\rho^2 \varepsilon}{4(n+8)K_d} \delta u$$

$$+ \varepsilon \delta u$$

$$= (\varepsilon - 2\varepsilon)\delta u = -\varepsilon \delta u$$

$$\frac{d}{dl}\begin{pmatrix} \delta t \\ \delta u \end{pmatrix} = \begin{pmatrix} 2 - \frac{n+2}{n+8}\varepsilon & a \\ 0 & -\varepsilon \end{pmatrix} \begin{pmatrix} \delta t \\ du \end{pmatrix}$$

Eigenvalues $\lambda_1 = 2 - \frac{n+2}{n+8}\varepsilon > 0$

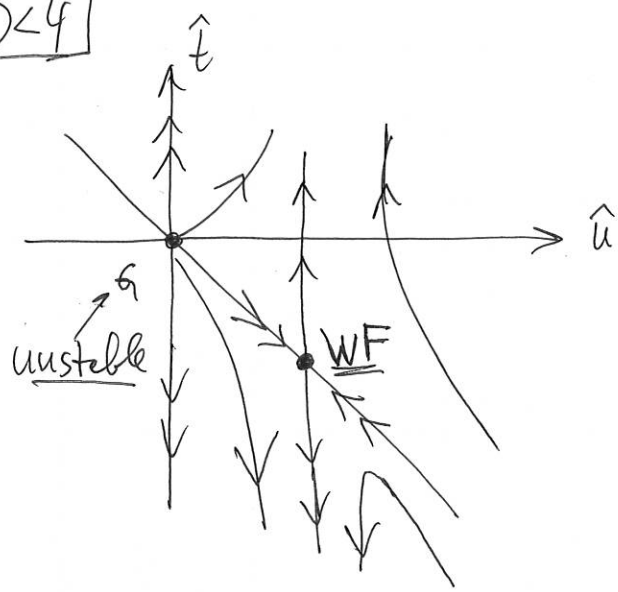
$\lambda_2 = -\varepsilon < 0$!!! stable direction

Eigenvalues $\lambda_1 = 2 - \frac{n+2}{n+8}\varepsilon$

$\lambda_2 = \varepsilon - 6b u^* = \varepsilon - 6b\frac{\varepsilon}{3b} = -\varepsilon$!!!

stable direction

$\boxed{D<4}$



Consequences: correlation length

under scaling $\xi(t,h) = b\,\xi(b^{y_t} t, b^{y_h} h)$

Choose $b^{y_t} t = 1 \Rightarrow \xi(t,h) = t^{-\frac{1}{y_t}}\,\xi\left(1, \frac{h}{t^{y_h/y_t}}\right)$

$b = t^{-1/y_t}$

$$\Rightarrow \xi \sim \frac{1}{(t)^{\frac{1}{y_t}}} = t^{-\nu}, \quad \nu = \frac{1}{y_t}$$

At the WF fixed point $y_t = 2 - \frac{n+2}{n+8}\varepsilon$

$$\Rightarrow \nu = \frac{1}{y_t} = \frac{1}{2 - \frac{n+2}{n+8}\varepsilon} \cong \frac{1}{2} + \frac{(n+2)}{2(n+8)}\varepsilon$$

For $n=1$ (scalar φ^4 theory)

$$\nu = \frac{1}{2} + \frac{3}{2\cdot 9}\varepsilon = \frac{1}{2} + \frac{\varepsilon}{6}$$

For $\varepsilon = 1$ $(d = 4-\varepsilon = 3!)$ $\nu = \frac{1}{2} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$

very close to experimental value $\nu \simeq 0.67$